

TONY SUI

0409 222 264 | tonysui50@gmail.com | Philadelphia, PA / Sydney, AU

EDUCATION

University of Pennsylvania

Philadelphia, PA

Exchange Semester (Wharton & SEAS); GPA: –

Jan 2026 – May 2026

- **Activities:** Wharton Undergraduate Entrepreneur Club (Conference Committee), Wharton Undergraduate Sports Business Club (Summit Planning Committee), Wharton Investment and Trading Group (Accelerator Program), Penn Claude Builders Club, Wharton AI Product Management Training Program, Wharton Music Business Club (Internals).

University of New South Wales

Sydney, Australia

Bachelor of Advanced Computer Science (Hons); WAM: 81.6

Expected Dec 2027

- **Honors:** AAA Scholarship, Leadership Scholarship, Best & Brightest (CSE) Scholarship, College Scholar, Talented Students Program.

Smith's Hill High School

Wollongong, Australia

HSC ATAR: 99.70 (Dux)

Graduated 2022

- Ranks: 7th in NSW for Economics (97/100), Ext 2 Maths (96/100), Chemistry (96/100).

EXPERIENCE

Wollongong City Council

Wollongong, Australia

Business Systems Analyst Cadet

Feb 2025 – Present

- Built and maintained SQL ETL pipelines, reducing end-to-end reporting times by 40%.
- Designed interactive dashboards for finance and operations teams to accelerate decision-making.
- Resolved 100+ service tickets leveraging OneCouncil data models and business process knowledge.

Private Tutor

Sydney, Australia

HSC Economics & STEM Tutor

Jan 2023 – Present

- Delivered 200+ hours of tutoring in Ext 1/2 Maths, Physics, and Economics; improved student results by avg. 20%.

PROJECTS

C++17 Limit Order Book & Market Simulator | C++, Low-Latency Systems

- Engineered a low-latency matching engine supporting Limit, Market, FOK, and IOC orders via price-time priority.
- Achieved **~100ns wire-to-wire latency** and **850k+ orders/sec** throughput by optimising critical execution paths.
- Developed a stochastic market simulator (Brownian motion) to validate fill dynamics (2.8ms avg fill time).
- Instrumented micro-benchmarks to identify memory allocation as the primary constraint on P99 tail latency ($<2\mu\text{s}$).

UDP Reliable Transport Protocol | Python, Network Engineering

- Designed a TCP-like reliable transport protocol over UDP featuring sliding windows, cumulative ACKs, and timers.
- Implemented fast retransmit, error detection, and a state machine for connection lifecycle management.

Video Sharing Platform | Next.js, TypeScript, AWS, Express.js

- Built a scalable platform with secure JWT authentication and optimised video delivery via CDN.

COMPETITIONS & AWARDS

Winner – Jane Street Mystery Planet Puzzle Challenge

2nd Place – Harvard Hackrare Hackathon (NORD Scholarship Recipient)

2nd Place – Citadel Securities Mock Trading Challenge

2nd Place – CSOC Hackathon (Built “Zwift” cycling app with route optimisation & tracking)

Participant – SIG Discovery Day Technology Stream (Selected 1 of 24 across ANZ)

Participant – Citadel Securities Trading Invitational Challenge

TECHNICAL SKILLS

Languages: C++17/20, Python, Java, C, JavaScript/TypeScript, SQL

Technologies: Git, Docker, Bash, AWS, Puppeteer, React, Linux, Jira

LEADERSHIP & ACTIVITIES

UNSW Societies: Peer Mentor (CSESOC & EngSoc), Subcommittee (Personal Investment & Medicine Societies).

Sports: 4th in State (Doubles Tennis, 2022), University of Pennsylvania Club Rugby.

High School Leadership: House Captain, Platinum Leadership Award, Da Vinci Decathlon State Awards.